Chapter 2 Teacher Key — Moving Objects Have Energy

> Combined answer keys: Moving Marbles · Practice · Chapter 2 Quiz.

Answer Key — Moving Marbles

- **Artifact:** `production/worksheets/02-ch2-moving-marbles-student.docx`
- **Source alignment:** AP 3.1 (On the Move); Reader Ch.2 pp. 7–12; TG L3.

Item	Correct/expected response	Acceptable alternatives	Reasoning	Common misconception	Scoring note	What NOT to penalize
PREDICT	a guess (target will move / stay)	any prediction	prediction = testable guess	wrong prediction = failure	1 pt	a "wrong" prediction
BEFORE	one marble rolling, target still	drawing/ description of still target	states initial state	—	1 pt	arrow style
TEST table — "What moved before?"	the rolled marble	"the moving marble"	identifies the mover	—	1 pt	—
TEST — "What changed after contact?"	the target started moving	"target slid/moved"	names the change	—	1 pt	—
TEST — "How did the target change?"	changed position (still → moving)	"it moved from here to there"	position changed = motion	"the marble had energy" (that's inference, not change)	1 pt	—
TEST — "Evidence the moving marble had energy"	the target changed position	"the target moved after being hit"	change = evidence of energy	energy is visible	2 pts	informal phrasing
MODEL (before/after)	two frames: still target → moved target, with arrows	any correct before/after	model shows cause→effect	—	2 pts	artistic quality
EXPLAIN (claim + evidence)	"The moving marble had energy of motion because it caused the target to	any claim + observed change	change is evidence	only moving objects have energy (inverse)	2 pts	full CER not required

Item	Correct/expected response	Acceptable alternatives	Reasoning	Common misconception	Scoring note	What NOT to penalize
	change position."					

Notes

- Measurement (target distance in cm) is OPTIONAL evidence — do not require exact displacement for the concept.
- A wrong prediction is NOT penalized if observation/explanation are correct.
- Do not award "the marble was fast" as evidence (that's Ch.3).

Answer Key — Chapter 2 Practice

- **Artifact:** `production/worksheets/02-ch2-moving-objects-have-energy-practice.docx`
- **Source alignment:** Reader Ch.2 pp. 7–12; TG L3; AP 3.1.

Part A — What is moving?

Scenario	Object in motion
A basketball falls through the hoop	the basketball
A marble rolls toward another marble	the rolled marble
Wind spins a turbine blade	the moving air (wind)
A shopping cart bumps a box	the shopping cart

Part B — What changed?

Scenario	Change caused
The basketball hits the net	the net moves (still → moving)
The marble hits the target marble	the target changes position
Wind spins the blades	the blades rotate (still → spinning)
The cart bumps the box	the box slides forward

Part C — Evidence (which observation shows the moving object had energy?)

Observation	Evidence?	Why
"The target marble changed position."	YES	a change happened caused by the mover
"The marble is round."	NO	shape is not a change caused by motion
"The block slid forward after the ball hit it."	YES	still → moving is a change
"I think the ball is heavy."	NO	opinion, not an observed change

Part D — Before / After (sample)

- Before: target block still. After: block moved. Arrows show the direction of the moving object and the target's motion.
- Accept any correct two-frame drawing with a clear change.

Part E — Explain (sample)

"A moving object has energy of motion. When it bumps another object and that object changes position, the change is evidence that the moving object had energy."

- Accept any response that (a) states a moving object has energy of motion, (b) names an observed change, (c) links the change to energy as evidence.
- Do NOT require speed/fast-slow language.

Answer Key — Chapter 2 Quiz

- **Artifact:** `production/quizzes/02-ch2-moving-objects-have-energy-quiz.docx`
- **Source alignment:** Reader Ch.2 pp. 7–12; TG L3; AP 3.1.

Item	Correct/expected response	Acceptable alternatives	Reasoning	Common misconception	Scoring note	What NOT to penalize
1	Motion is the process of an object changing position	—	Reader p.7	motion = "speed"	1 pt	—
2	A moving object has energy of motion	—	Reader p.9	energy of motion = "speed"	1 pt	—

Item	Correct/expected response	Acceptable alternatives	Reasoning	Common misconception	Scoring note	What NOT to penalize
3	(scenario) moving object = the rolled marble; change = target moved	correct mover + change	Reader p.8	—	2 pts (1 each)	—
4	Energy is the ability to cause change	—	Reader p.1 (Ch.1)	energy as substance	1 pt	—
5	before/after diagram showing the change	any correct two-frame change	AP 3.1	—	2 pts	arrow style
6	"The target moved" = observation; "The marble had energy" = inference	correct classification	TG L3	treating inference as observation	2 pts (1 each)	—
7	(short) "The moving object caused the target to change position; the change is evidence of energy of motion."	claim + change + evidence	Reader p.9	—	2 pts	informal phrasing
8	A still object (book on table) does NOT show energy of motion because nothing is moving and no change happens	correct non-example reasoning	ADAPTED	—	1 pt	—

Out-of-scope confirmation

- No speed/fast-slow · no ramp height · no mass · no transfer/transformation · no collisions · no kinetic/potential · no engineering.

Total

~13 points (adjust denominator per school grading policy — see OPEN_QUESTIONS).